

HIV Analysis

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```
rm(list = ls())
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.2.0      v readr      2.1.6
```

```
## v forcats    1.0.1      v stringr    1.6.0
```

```
## v ggplot2    4.0.2      v tibble     3.3.1
```

```
## v lubridate  1.9.5      v tidyr      1.3.2
```

```
## v purrr      1.2.1
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(janitor)
```

```
##
```

```
## Attaching package: 'janitor'
```

```
##
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
##      chisq.test, fisher.test
```

```
deaths <- read_csv("Data/no_of_deaths_by_country_clean.csv")
```

```
## Rows: 510 Columns: 7
```

```
## -- Column specification -----
```

```
## Delimiter: ","
```

```
## chr (3): Country, Count, WHO Region
```

```
## dbl (4): Year, Count_median, Count_min, Count_max
```

```
##
```

```
## i Use `spec()` to retrieve the full column specification for this data.
```

```
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
hiv <- read_csv("Data/no_of_people_living_with_hiv_by_country_clean.csv")
```

```
## Rows: 680 Columns: 7
```

```
## -- Column specification -----
```

```
## Delimiter: ","
```

```
## chr (3): Country, Count, WHO Region
```

```
## dbl (4): Year, Count_median, Count_min, Count_max
```

```
##
```

```
## i Use `spec()` to retrieve the full column specification for this data.
```

```
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```

art <- read_csv("Data/art_coverage_by_country_clean.csv")

## Rows: 170 Columns: 11
## -- Column specification -----
## Delimiter: ","
## chr (5): Country, Reported number of people receiving ART, Estimated number ...
## dbl (6): Estimated number of people living with HIV_median, Estimated number...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.

deaths_clean <- deaths |>
  clean_names() |>
  group_by(country) |>
  summarise(
    deaths = first(count_median),
    .groups = "drop"
  )

hiv_clean <- hiv |>
  clean_names() |>
  group_by(country) |>
  summarise(
    hiv_population = first(count_median),
    .groups = "drop"
  )

art_clean <- art |>
  clean_names() |>
  select(
    country,
    art_coverage = estimated_art_coverage_among_people_living_with_hiv_percent_median
  )

final_df <- deaths_clean |>
  inner_join(hiv_clean, by = "country") |>
  inner_join(art_clean, by = "country")

write_csv(final_df, "Data/final_df.csv")

glimpse(final_df)

## Rows: 170
## Columns: 4
## $ country      <chr> "Afghanistan", "Albania", "Algeria", "Angola", "Argenti~
## $ deaths       <dbl> 500, NA, 200, 14000, 1700, 200, 200, NA, NA, 200, NA, 5~
## $ hiv_population <dbl> 7200, NA, 16000, 330000, 140000, 3500, 28000, NA, NA, 6~
## $ art_coverage  <dbl> 13, NA, 81, 27, 61, 53, 83, NA, NA, 52, NA, 22, 50, 59,~

```

Description

I created a merged, final dataset to use for analysis with cleaned and standardized columns (contains 170 countries and 4 variables). Each row represents a single country. The dataset was constructed by merging three country-level datasets sourced from WHO-based HIV/AIDS indicators available on Kaggle. All variables are continuous numerical estimates, and represent median values reported in the original datasets. The variables included in the final dataset are: - HIV/AIDS deaths (deaths) - Number of people living with HIV

(hiv_population) - ART coverage among people living with HIV (art_coverage)

```
colSums(is.na(final_df))
```

```
##      country      deaths hiv_population  art_coverage
##           0           37           32           34
```

```
colMeans(is.na(final_df))
```

```
##      country      deaths hiv_population  art_coverage
## 0.0000000 0.2176471 0.1882353 0.2000000
```

Description

The dataset contains missing values across all three main variables: HIV/AIDS deaths (37 missing, ~21.8%), number of people living with HIV (32 missing, ~18.8%), and ART coverage (34 missing, 20%); no missing data for countries. This is most likely due to incomplete reporting in certain countries within the original WHO-based dataset (real-world data is often incomplete). Missing data was not cleaned or removed, just be wary of it and decide what to do with it in the next stages (if anything).

```
final_df |>
  arrange(desc(deaths)) |>
  slice(1:10)
```

```
## # A tibble: 10 x 4
##   country      deaths hiv_population art_coverage
##   <chr>      <dbl>      <dbl>      <dbl>
## 1 South Africa  71000      7700000      62
## 2 Mozambique   54000      2200000      56
## 3 Nigeria      53000      1900000      53
## 4 Indonesia    38000      640000      17
## 5 Kenya      25000      1600000      68
## 6 United Republic of Tanzania 24000      1600000      71
## 7 Uganda       23000      1400000      72
## 8 Zimbabwe     22000      1300000      88
## 9 Cameroon     18000      540000      52
## 10 Thailand    18000      480000      75
```

Description

Shows top 10 countries with the most HIV/AIDS deaths.

```
final_df |>
  filter(!is.na(deaths)) |>
  arrange(deaths) |>
  slice(1:10)
```

```
## # A tibble: 10 x 4
##   country      deaths hiv_population art_coverage
##   <chr>      <dbl>      <dbl>      <dbl>
## 1 Barbados     100        3000      50
## 2 Bhutan        100        1300      37
## 3 Bosnia and Herzegovina 100         500      67
## 4 Bulgaria      100        3500      41
## 5 Cabo Verde    100        2400      89
## 6 Comoros        100         200      79
## 7 Croatia        100        1600      75
```

##	8	Czechia	100	4400	60
##	9	Denmark	100	6200	89
##	10	Estonia	100	7400	59

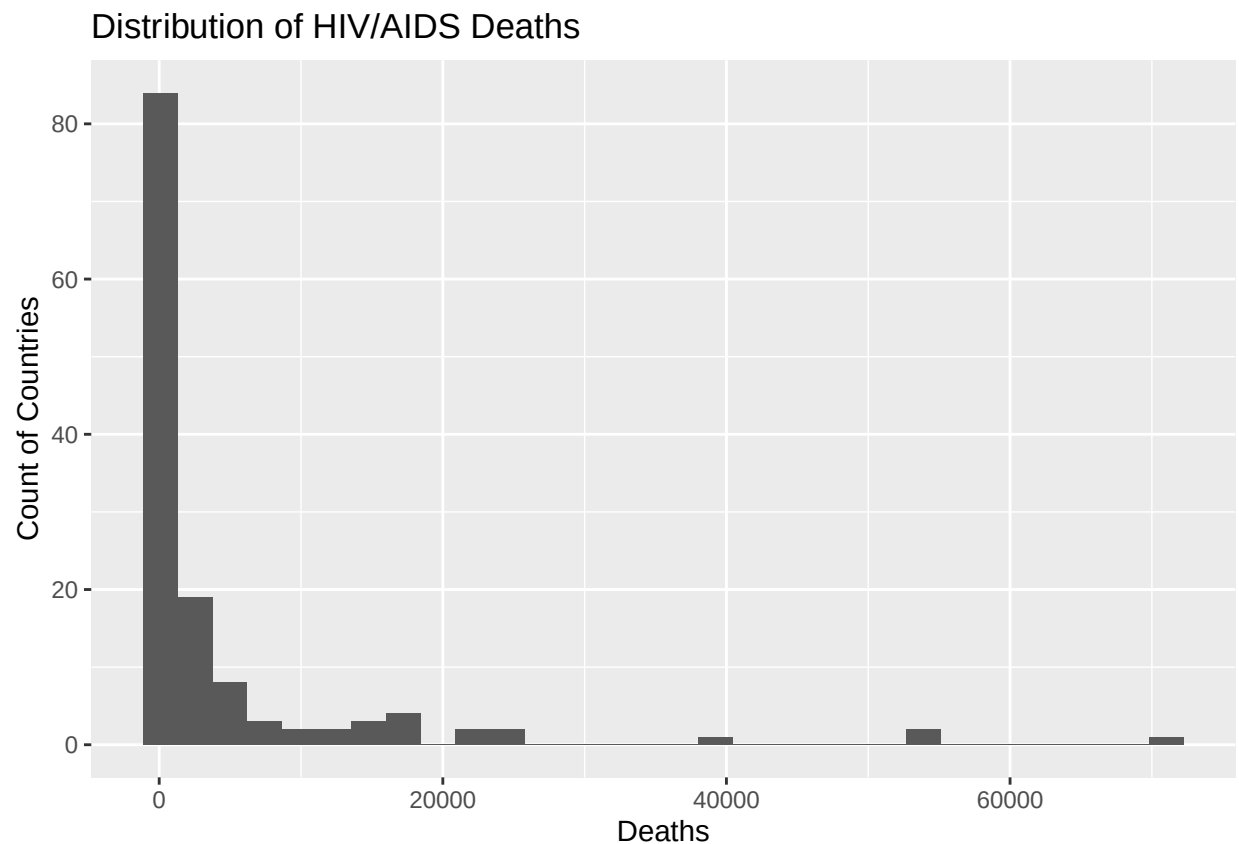
Description

Shows top 10 countries with the least HIV/AIDS deaths.

```
ggplot(final_df, aes(x = deaths)) +
  geom_histogram(bins = 30) +
  labs(title = "Distribution of HIV/AIDS Deaths", x = "Deaths", y = "Count of Countries")
```

Warning: Removed 37 rows containing non-finite outside the scale range

(`stat\_bin()`).



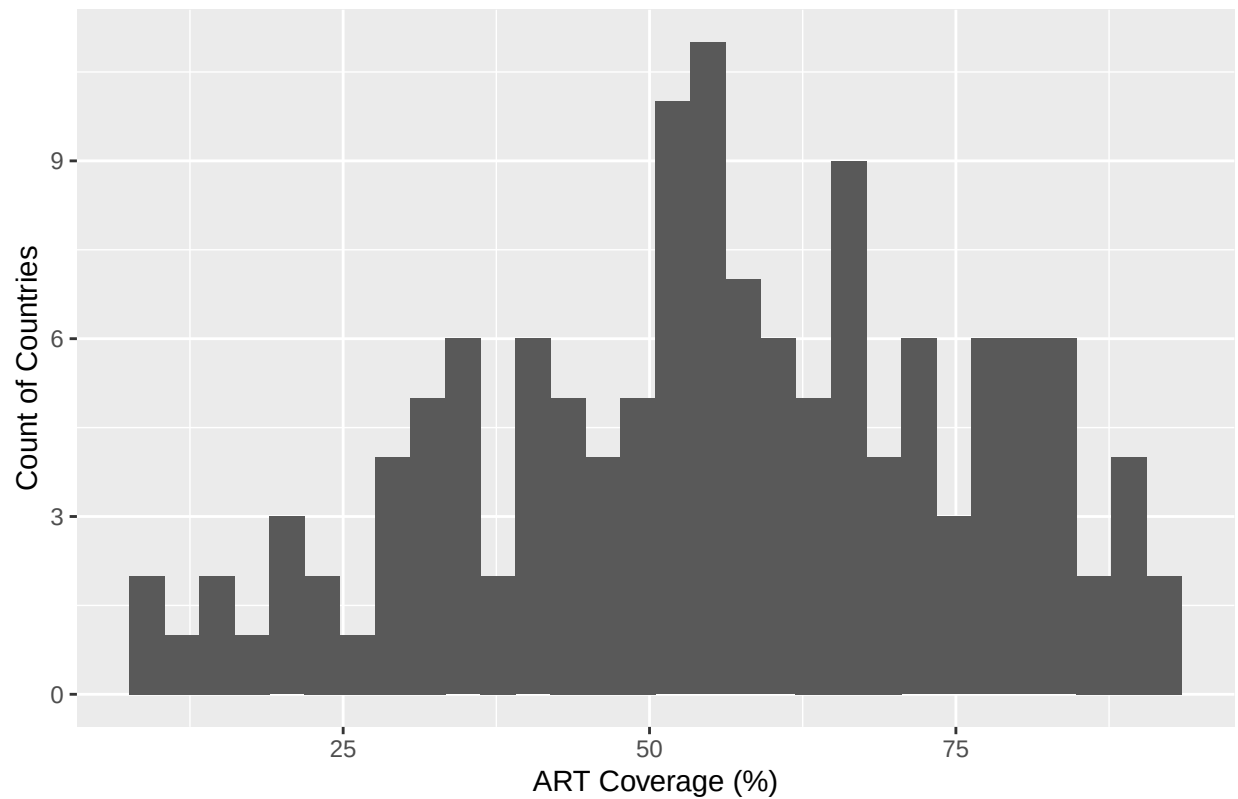
Description Shows the distribution of deaths across countries.

```
ggplot(final_df, aes(x = art_coverage)) +
  geom_histogram(bins = 30) +
  labs(title = "Distribution of ART Coverage", x = "ART Coverage (%)", y = "Count of Countries")
```

Warning: Removed 34 rows containing non-finite outside the scale range

(`stat\_bin()`).

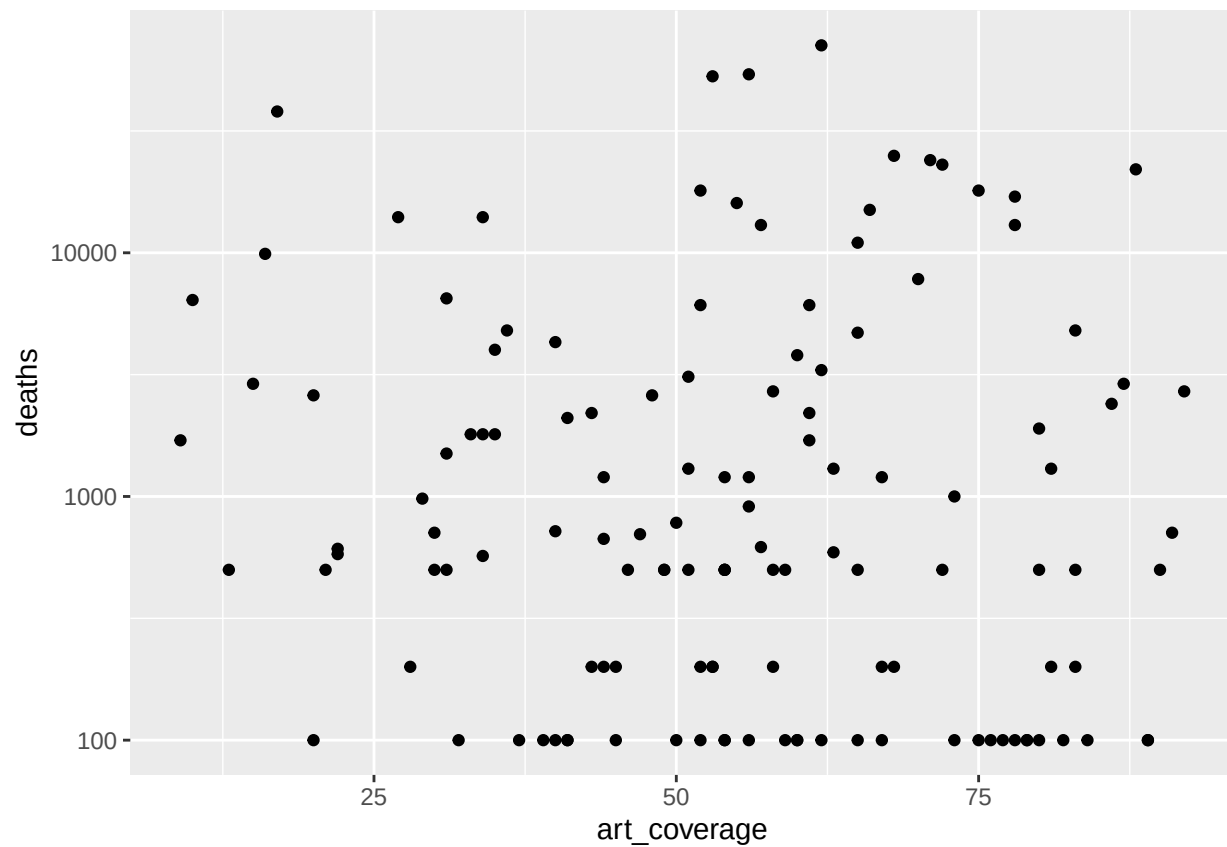
Distribution of ART Coverage



Description Shows the distribution of treatment access (ART coverage %) across countries.

```
ggplot(final_df, aes(x = art_coverage, y = deaths)) +  
  geom_point() +  
  scale_y_log10()
```

Warning: Removed 39 rows containing missing values or values outside the scale range
(`geom\_point()`).



Description Shows the relationship between ART coverage (%) and HIV/AIDS deaths across countries, where each point represents a country. Note that I used log scaling for the y-axis to account for scaling issues and compression of the graph.